

Complements help computers subtract by using addition instead of borrowing.
There are 2 types of complements:

1) Diminished Radix Complement ("Just Flip")

Binary DRC is called 1's complement

Decimal DRC is called 9's complement

Octal DRC is called 7's complement

Hexadecimal DRC is called 15's complement

To get the DRC, subtract each digit from the maximum allowed value in the base.

Ex: 9's comp of $(546700)_{10}$

$$\begin{array}{r} 999999 \\ - 546700 \\ \hline 453299 \end{array} \quad \left. \begin{array}{l} \text{Because we have base 10, the maximum value allowed for each} \\ \text{digit is 9. So the 9's complement of 546700 is} \end{array} \right\} 453299$$

Ex: 1's comp of $(1011000)_2$

$$\begin{array}{r} 111111 \\ - 101100 \\ \hline 010011 \end{array} \quad \left. \begin{array}{l} \text{Because we have base 2, the maximum value allowed for each} \\ \text{digit is 1. So the 1's complement of 1011000 is} \end{array} \right\} 010011$$

1's complement shortcut: make all 0s into 1s and
make all 1s into 0s.

Ex: 1's comp of $(11010)_2 = (00101)_2$

2) Radix Complement ("Flip + 1")

Binary RC is called 2's complement

Decimal RC is called 10's complement

Octal RC is called 8's complement

Hexadecimal RC is called 16's complement

To get the RC:

1) Find the DRC (1s\9s\7s\15s comp)

2) Add 1 to the DRC.

Ex: 10s comp of $(2389)_{10}$

$$\begin{array}{r} 9999 \\ - 2389 \\ \hline 7610 \end{array} \quad \begin{array}{r} 7610 \\ + 1 \\ \hline 7611 \end{array}$$

So the 10s comp of 2389 is 7611

Ex: 2s comp of $(101100)_2$

$$\begin{array}{r} 010011 \\ + 1 \\ \hline 010100 \end{array}$$

Shortcut

So the 2s comp of 101100 is 010100.

Ex: 16s comp of $(3DF2)_{16}$

$$\begin{array}{r} \text{FFFF} \\ - 3DF2 \\ \hline \end{array} \rightarrow \begin{array}{cccc} 15 & 15 & 15 & 15 \\ - 3 & 13 & 15 & 2 \end{array} \rightarrow \begin{array}{cccc} \text{DRC (15's)} \\ 12 & 2 & 0 & 13 \end{array} \rightarrow \begin{array}{cccc} & + 1 \\ 12 & 2 & 0 & 13 \\ & & & + 1 \end{array} \rightarrow C20E$$

To solve $M - N$, there are 2 ways to do it

"Radix Complement" Method

- 1) Make sure both M and N have the same amount of digits. If not, add 0s to the left side.
- 2) Take the RC of N (2s\10s\8s\16s)
- 3) Add it to M
- 4) If the addition results in a left carry-over, the answer to the subtraction will be positive.
Discard the carry over.
- 5) Otherwise, the answer to the subtraction will be negative.

$$\begin{array}{l} \overset{m}{1010100} - \overset{n}{1000011} \end{array} \left. \begin{array}{l} n \text{ has 7 bits} \\ m \text{ has 7 bits} \end{array} \right\}$$

- 1) $1010100 - 1000011 \checkmark$
- 2) 2s comp of 1000011 is 0111101
- 3) $\begin{array}{r} 1010100 \\ + 0111101 \\ \hline 10010001 \end{array} \rightarrow 0010001$
- 4) Carry over exists. The answer is positive.

"Diminished Radix Complement" Method

- 1) Make sure both M and N have the same amount of digits. If not, add 0s to the left side.
- 2) Take the DRC of N (1s\9s\7s\15s)
- 3) Add it to M
- 4) If the addition results in a left carry-over, the answer to the subtraction will be positive.
add the carry-over to the sum.
- 5) Otherwise, the answer to the subtraction will be negative.

$$\begin{array}{l} \overset{m}{10101} - \overset{n}{0101} \end{array} \left. \begin{array}{l} n \text{ has 4 bits} \\ m \text{ has 5 bits} \end{array} \right\}$$

- 1) $10101 - 00101$
- 2) 1's comp of 00101 is 11010
- 3) $\begin{array}{r} 10101 \\ + 11010 \\ \hline 101111 \end{array} \leftarrow \text{carry over exists, positive.}$
- 4) $\begin{array}{r} 01111 \\ + 1 \\ \hline 10000 \end{array}$